

SntB triggers the antioxidant pathways to regulate development and aflatoxin biosynthesis in *Aspergillus flavus*

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Abstract

The epigenetic reader SntB was identified as an important transcriptional regulator of growth, development, and secondary metabolite synthesis in *Aspergillus flavus*. However, the underlying molecular mechanism is still unclear. In this study, *sntB* gene deletion ($\Delta sntB$), complementary (Com-*sntB*), and HA tag fused to *sntB* (*sntB*-HA) strains were constructed by using the homologous recombination method, respectively. Our results revealed that deletion of *sntB* inhibited the processes of mycelia growth, conidial production, sclerotia formation, aflatoxin synthesis, and ability to colonize host compared to wild type (WT), and the defective phenotype of knockout strain $\Delta sntB$ could be restored in its complementary strain Com-*sntB*. Chromatin immunoprecipitation sequencing (ChIP-seq) of *sntB*-HA and WT and RNA sequencing (RNA-seq) of $\Delta sntB$ and WT strains revealed that SntB played key roles in oxidative stress response of *A. flavus*. The function of *catC* (encode a catalase) gene was further analyzed based on the integration results of ChIP-seq and RNA-seq. In $\Delta sntB$ strain, the relative expression level of *catC* was significantly higher than in WT strain, while a secretory lipase encoding gene (G4B84_008359) was down-regulated. Under the stress of oxidant menadione sodium bisulfite (MSB), the deletion of *sntB* obvious down-regulated the expression level of *catC*. After deletion of *catC* gene, the mycelia growth, conidial production, and sclerotia formation were inhibited, while ROS level and aflatoxin production were increased compared to the WT strain. Results also showed that the inhibition rate of MSB to $\Delta catC$ strain was significantly lower than that of WT group and AFB1 yield of the $\Delta catC$ strain was significantly decreased than that of WT strain under the stress of MSB. Our study revealed the potential machinery that SntB regulated fungal morphogenesis, mycotoxin anabolism, and fungal virulence through the axle of from SntB to fungal virulence and mycotoxin bio-synthesis, i.e. H3K36me3 modification-SntB-Peroxisomes-Lipid hydrolysis-fungal virulence and mycotoxin bio-synthesis. The results of the study shed light into the SntB mediated transcript regulation pathway of fungal mycotoxin anabolism and virulence, which provided potential strategy for control the contamination of *A. flavus* and its aflatoxins.

eLife assessment

In this **useful** study, the authors investigate the regulatory mechanisms related to toxin production and pathogenicity in *Aspergillus flavus*. Their observations indicate that the SntB protein regulates morphogenesis, aflatoxin biosynthesis, and the oxidative stress response. In general, the data supporting the conclusions are **solid** but could be strengthened further through additional analyses of CHIP-seq data.

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1. Introduction

Aspergillus flavus is one of the common asexual species, a saprophytic fungus and the second largest pathogenic fungus after *Aspergillus fumigatus*, widely distributed in soil, air, water, plants, and agricultural products in nature [1]. Aflatoxins produced by *A. flavus* have strong toxicity, and is extremely harmful to human society. Animal and human health can be negatively affected by aflatoxins, which are carcinogenic, teratogenic, and mutagenic [2]. Among aflatoxins, AFB1 is the most frequently occurring and the most toxic and carcinogenic, which is converted to AFB1-8 and 9-epoxide in the liver and formed adducts with the guanine base of DNA and thus results in acute and chronic diseases in both human and household animals [3]. According to the Food and Agriculture Organization of the United Nations, 25% of the food crops in the world are contaminated with aflatoxins [4], which are often detected in grains, nuts, and spices [5, 6]. It is urgent to control the contamination of *A. flavus* and its main mycotoxin, AFB1.

In recent decades, the biosynthetic pathway of aflatoxins was investigated in detail benefit from the sequence of *A. flavus* genome [7]. This pathway consists of a complex set of enzymatic reactions [8–11]. In general, these enzymes are encoded by clusters of genes, which are regulated by cluster-specific genes: *aflR* and *aflS* [12, 13]. The initial stage of aflatoxins biosynthesis is catalyzed by polyketide synthase (PKSA) to form the polyketone backbone [14]. The synthesis of aflatoxins is additionally influenced by environmental stimuli such as pH, light exposure, nutrient availability, and the response to oxidative stress, potentially leading to the alteration of gene expression related to toxin biosynthesis [15–17]. Besides the biosynthetic pathway and its internal gene regulation, protein post-translational modifications (PTMs), an important mean of epigenetics, represent an important role in the regulation of aflatoxins synthesis, including 2-hydroxyisobutyrylation, succinylation, acetylation, and methylation [18–24], in which Snt2 (also called *sntB*, an epigenetic reader) is deeply involved. Despite advancements in the field, our understanding of the molecular mechanisms of aflatoxin production in *A. flavus* is still fragmentary.

The epigenetic reader encoded by *sntB* in *A. nidulans* was identified as a transcriptional regulator of the sterigmatocystin biosynthetic gene cluster and deletion of *sntB* gene in *A. flavus* results in loss of aflatoxin production [25], increasing global levels of H3K9K14 acetylation and impairing several developmental processes [20]. The homolog gene in yeast, SNTB coordinates the transcriptional response to hydrogen peroxide stress [26, 27]. In *Penicillium expansum*, SntB regulated the development, patulin and citrinin production, and virulence on apples [28]. In *A. nidulans*, SntB combined with a H3K4 histone demethylase KdmB, a cohesin acetyltransferase (EcoA), and a histone deacetylase (RpdA) to form a chromatin binding complex (KERS) and bound to regulatory genes and coordinated fungal development with mycotoxin synthesis [29]. In *A. flavus*, the KERS complex also consists of KdmB, RpdA, EcoA, and SntB and plays a key role in the

fungus development and secondary metabolites metabolism [30]. *sntB* also regulated the virulence in *Fusarium oxysporum*, and respiration in *F. oxysporum* and *Neurospora crassa* [31, 32].

However, the specific regulatory mechanism of *sntB* in *A. flavus* remains unclear. In this study, we identified the regulatory network of *sntB* by Chromatin immunoprecipitation sequencing (ChIP-seq) and RNA sequencing (RNA-seq), which shed light on its impact on fungal biology.

Experimental Procedures

A. flavus strains, media, and culture conditions

A. flavus $\Delta ku70 \Delta pyrG$ was used as a host strain, for genetic manipulations. All strains used in this study are listed in the Table 1. potato dextrose agar (PDA, 39 g/L, BD, Difco, Franklin, NJ, USA), complete medium (CM, 6 g/L tryptone, 6 g/L yeast extract, 10 g/L glucose), and potato dextrose (PDB, 24 g/L, BD, Difco, Franklin, NJ, USA) were used for mycelial growth and sporulation determination, sclerotia production, and mycotoxin production analysis, respectively. All experiments were technically repeated three times and biologically repeated three times.

The construction of mutant strains

All mutant strains, including *sntB* and *catC* gene knock-out strain ($\Delta sntB$ and $\Delta catC$), the complementation strain for the $\Delta sntB$ strain (Com-*sntB*), and HA tag fused to *sntB* strain (*sntB*-HA), were constructed following the protocol of homologous recombination [34] and the detail protocol was as described in our previous study [35]. The related primers were listed in Table 2. The constructed strains were confirmed by diagnostic PCRs and by southern blot [36]. The construction of *sntB*-HA was further determined by western blot with anti-HA antibody as described previously [35].

Phenotypic analysis and aflatoxin analysis

The spores (10^7 conidia/mL) of wild type (WT), $\Delta sntB$, and Com-*sntB* strains were used. The details of experiment were according to our previous study [37]. Hyphal septum was stained according to the described method [38]. Each fungal strain was evaluated on four plates, and each experiment was repeated three times.

Fungal colonization on crop kernels

According to our previous experimental protocol [37], the colonizing ability of WT, $\Delta sntB$, and Com-*sntB* fungal strains on peanut and corn kernels was analyzed. The crop kernels were disinfected with 0.05% sodium hypochlorite and soaked for 30 min in a solution containing 10^5 conidia/mL fungal spores. Afterwards, the seeds were placed in a petri dish and cultured at 29°C for 6 d. Finally, the number of conidia was calculated and AFB1 product was analyzed by TLC.

Animal invasion assay

The animal invasion assay using silkworms was conducted according to a previous study [34, 39]. Silkworms (*Bombyx mori*) were randomly separated into four groups (10 larvae/group) when silkworm larva reach about 1 g in weight. Each silkworm was injected with 5 μ L saline, or 5 μ L conidial suspension (10^6 spores/mL) from WT, $\Delta sntB$, and Com-*sntB* strains. The survival rate of silkworms was calculated. Dead silkworms were transferred into fresh 9 cm Petri dishes and inoculated for 5 d in the dark. The conidia number and AFB1 production from each group were measured.

Table 1.

***A. flavus* strains used in this study**

Strain name	Related genotype	Source
<i>A. flavus</i> CA14	$\Delta ku70, \Delta pyrG$	Kindly presented from Prof. Chang ^[33]
Wild-type (WT)	$\Delta ku70, \Delta pyrG::Afp yrG$	This study
$\Delta sntB$	$\Delta ku70, \Delta sntB::Afp yrG$	This study
Com- <i>sntB</i>	$\Delta ku70, \Delta pyrG; \Delta sntB::Afp yrG::sntB$	This study
$\Delta catC$	$\Delta ku70, \Delta catC::Afp yrG$	This study

Primer name	Sequence (5' → 3')	Fragment amplified
<i>sntB</i> -p1	CTTCTCGAATTCCCCTTCATGACACTCTCC	
<i>sntB</i> -p2	GCTAAATCAGGATGGGTGGAGGGTGAC	
<i>sntB</i> -p3	GCATCTCCTTTGTGTTGTTTGGACCGTGT	
<i>sntB</i> -p4	CAACCAACCACTGACGTGACCAC	
<i>sntB</i> -p5	GTCACCCTCCAACCCATCCTGATTAGCGCC TCAAACAATGCTCTTCACCC	For the construction of knockout strain
<i>sntB</i> -p6	ACACGGTCCAAACAACACAAAGGAGATGCG TCTGAGAGGAGGCACTGATGC	
<i>sntB</i> -p7	TAGATCACCCAGCGGGCCACAA	
<i>sntB</i> -p8	GACTCAAATGGAAATCCCGTCGTGCC	
<i>sntB</i> -F	GCAGCAACACCACGTGAGGCCCAATTC	a fragment from <i>sntB</i> ORF
<i>sntB</i> -R	CCAGGTCACAGGGCATAGAACACACTCGTA	
P1020-F	ATCGGCAATACCGTCCAGAAGC	Verify the fragment of <i>Afp_{yrG}</i>
P801-R	CAGGAGTTCTCGGGTTGTGC	

Table 2.

Primers used for strain construction in this study

<i>sntB</i> -C-p2	GGGTGAAGAGCATTGTTTGAGGCCCTCCAA CCTTACTCCGTACACAATTCTAG	
<i>sntB</i> -C-p3	GCATCAGTGCCTCCTCTCAGACGGCACGAC GGGATTTCATTTGAGTC	For the construction of complementary strain of <i>sntB</i>
<i>sntB</i> -C-p4	CCAATTTCTGATGATTGTGATGTGTGTCC	
SC-P-F	GCCTCAAACAATGCTCTTCACCC	
SC-P-R	GTCTGAGAGGAGGCACTGATGC	
<i>catC</i> -P1	CTTGAGACGCAGGACGAA	For the construction of knockout strain of <i>catC</i>
<i>catC</i> -P2	GGGTGAAGAGCATTGTTTGAGGCTGATGTG GGTTGTATGAATG	
<i>catC</i> -P3	GCATCAGTGCCTCCTCTCAGACTGGATGCGG GTGAATACTG	
<i>catC</i> -P4	ACAAGCTGTCATGCGTGG	
<i>catC</i> -P5	TGGGAGTCTCGAACACAC	
<i>catC</i> -P6	GAAAACCCCGCAACAGAC	
<i>catC</i> -F	AGCCTATTTTCGGACCCCT	
<i>catC</i> -R	CAGTCTCCTTTCGGCATC	

Table 2. (continued)

RNA-seq analysis

To reveal the potential complex regulatory network of the *sntB*, RNA-seq analysis was carried out on the WT and $\Delta sntB$ strains. RNA-seq analysis was carried out by APPLIED PROTEIN TECHNOLOGY, Shanghai (www.apptbiotech.com) [40]. Data processing was according to a previous study [41]. Differentially expressed gene (DEGs) were assigned as genes with $|\log_2\text{FoldChange}| > 1$ and adjusted $P\text{-adj} < 0.05$. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways were used to analyze the functions of DEGs.

ChIP-seq and data analysis

ChIP-seq analysis was carried out on the WT and *sntB*-HA strains. The conidia ($10^4/\text{mL}$) for each strain were inoculated in 100 mL PDB shaking at 180 rpm under 29°C for 72h, and subjected to ChIP-seq analysis by Wuhan IGENEBOOK Biotechnology Co.,Ltd (<http://www.igenebook.com>). ChIP experiment was according to a previous study [34]. Raw sequencing with low-quality reads discarded and reads contaminated with adaptor sequences trimmed were filtered by Trimmomatic (v0.36) [42]. The clean reads were mapped to the reference genome of *A. flavus* by Burrows-Wheeler Alignment tool (BWA, version 0.7.15) [43]. MACS2 (v2.1.1) and Bedtools (v2.25.0) were used for peak calling and peak annotation, respectively. Differential binding peaks were identified by Fisher's test with q value < 0.05 . HOMER (version3) was used to predict motif occurrence within peaks with default settings for a maximum motif length of 12 base pairs [44]. Genes less than 2000 bp away were associated to the corresponding peak. GO and KEGG enrichment analyses of annotated genes were implemented in EasyGO [45] and KOBAS (v2.1.1) [46], with a corrected P -value cutoff of 0.05.

Quantitative RT-PCR (qRT-PCR) analysis

The fungal spores ($10^6/\text{mL}$) were cultured in PDB medium for 48 h, and then mycelium was ground into powder with liquid nitrogen. Total RNA was prepared by TRIpure total RNA Extraction Reagent (Bestek, China) according to the protocol used by Zhang [47]. qRT-PCR was performed according to previously described [48], and the primers were shown in Table 3.

Oxidative stress assays

To evaluate the role of *sntB* in fungal resistance to oxidative stress, a series concentration (0, 0.12, 0.24, and 0.36 mM) of menadione sodium bisulfite (MSB) were added to the medium. 10^6 fungal spores for each strain were inoculated on the medium and cultured in dark at 37°C. The diameters of colonies were measured 3 d after inoculation and the inhibition rate was calculated as previously described [35]. The AFB1 product was analyzed by TLC after the strains were cultured in YES medium in dark at 29°C for 7 d.

Reactive oxygen species assay

With the instructions provided in the user's manual, the intracellular ROS production was measured using a ROS assay kit (S0033S, Beyotime Institute of Biotechnology, China). After harvest, the mycelia were incubated with 10 μM 6-carboxy-2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) and 50 g/mL Rosup for 30 min. With SpectraMax Imaging Cytometer (Molecular Devices, Sunnyvale, Calif.) at emission wavelength of 525 nm and excitation wavelength of 488 nm, fluorescence signals of intracellular ROS production were acquired.

Statistical analysis

All data in this study were expressed as mean $\pm$ standard deviation. The statistical analysis was performed using the software GraphPad Prism8 (GraphPad Software, La Jolla, CA, USA). The difference was considered to be statistically significant when $P < 0.05$.

Primer name	Sequence (5' → 3')	Fragment amplified
<i>sntB-qF</i>	ACTCATCAGAGCCCCTATGGGCC AGTC	
<i>sntB-qR</i>	GCACTAAGAACGCGATCGACAG AATAGACAC	
<i>catC-qF</i>	GAAAGAGTTGTCCATGCCA	
<i>catC-qR</i>	CAGAAAACGGGTGTGTGAT	
<i>brlA-F</i>	GCCTCCAGCGTCAACCTTC	

Table 3.

Primers used for RT-qPCR in this study

<i>brlA</i> -R	TCTCTTCAAATGCTCTTGCCTC	
<i>abaA</i> -F	TCTTCGGTTGATGGATGATTTC	
<i>abaA</i> -R	CCGTTGGGAGGCTGGGT	
<i>nsdC</i> -F	GCCAGACTTGCCAATCAC	
<i>nsdC</i> -R	CATCCACCTTGCCCTTTA	
<i>nsdD</i> -F	GGACTTGCGGGTCTGTGCTA	
<i>nsdD</i> -R	AGAACGCTGGGTCTGGTGC	Sclerotium related genes
<i>sclR</i> -F	CAATGAGCCTATGGGAGTGG	
<i>sclR</i> -R	ATCTTCGCCCCGAGTG GTT	
<i>aflC</i> -F	GTGGTGGTTGCCAATGCG	
<i>aflC</i> -R	CTGAAACAGTAGGACGGGAGC	
<i>aflP</i> -F	CGATGTCTATCTTCTCCGATCTAT TC	Toxin synthesis structure genes
<i>aflP</i> -R	TCTCAGTCTCCAGTCTATTATCTA CC	
<i>aflO</i> -F	CTTTCGGCAGTGACCTAACC	
<i>aflO</i> -R	TCTTGA ACTATAAGGCGACCA	
<i>aflR</i> -F	AAAGCACCTGTCTTCCCTAAC	
<i>aflR</i> -R	GAAGAGGTGGGTCAGTGTTTGT AG	Toxin synthesis regulatory genes
<i>aflS</i> -F	GCTCAGACTGACCGCCGCTC	
<i>aflS</i> -R	GCTCAGACTGACCGCCGCTC	
<i>18S rRNA</i> -F	CTGAAGACTAACTACTGCGAAA GC	
<i>18S rRNA</i> -R	GAGCGGGTCATCATAGAAACAC	RNA extraction quality testing
<i>β-tublin</i> -F	TTGAGCCCTACAACGCCACT	
<i>β-tublin</i> -R	TGGTTCAGGTCACCGTAAGAGG	
<i>actin</i> -F	ACGGTGTCGTCACAAACTGG	
<i>actin</i> -R	CGGTTGGACTTAGGGTTGATAG	Internal reference gene

Table 3. (continued)

Results

The phenotype of SntB in *A. flavus*

The role of SntB in *A. flavus* has been previously characterized by analyzing both $\Delta sntB$ and overexpression *sntB* genetic mutants [20]. To further investigate the intrinsic mechanism of this regulator on the development and aflatoxin biosynthesis in *A. flavus*, the *sntB* deletion strain ($\Delta sntB$) and the complementary strain (Com-*sntB*) were constructed by the method of homologous recombination and verified by diagnostic PCR and southern blotting (Figure S1). The expression levels of *sntB* in WT, $\Delta sntB$, and Com-*sntB* strains was further detected by qRT-PCR and the result showed that the expression of *sntB* was absent in the gene-deletion strains, and it fully recovered in the Com-*sntB* (Figure S1), which reflected that the *sntB* and Com-*sntB* had been constructed, and could be used in the subsequent experiments of this study. The phenotype analysis of this study revealed that the deletion of *sntB* gene significantly inhibited the growth of mycelium, hyphae morphology, the length of fungal cell (between two adjacent septa), the number of conidiation, sclerotium formation, and aflatoxin biosynthesis, while the above phenotypes of both development and mycotoxin bio-synthesis were fully recovered in the Com-*sntB* strain (Figure 1). To reveal the signaling pathways of *sntB* in conidiation, sclerotium formation, and aflatoxin biosynthesis, qRT-PCR analysis was performed to assess the expression levels of sporulation related transcriptional factor genes, *steA*, *wetA*, *fluG*, and *veA*, sclerotia formation related transcriptional factor genes, *nsdC*, *nsdD*, and *sclR* [49, 50], and the AFs synthesis gene cluster structural genes *aflC*, *aflR*, and *aflP*, and the main regulatory gene *aflR* and *aflS*. As shown in Figure S2, the relative expression levels of these genes were significantly lower in the $\Delta sntB$ strain compared to that of the WT strain, and recovered in the Com-*sntB* strain. These results indicated that SntB regulates the conidiation, sclerotium formation, and aflatoxin biosynthesis by the canonical signaling pathways mediated by these regulators.

SntB plays important role in virulence of *A. flavus* to both plant and animal hosts

In order to explore the effect of SntB on the fungal colonization ability, peanut and maize kernels were infected with spore solution of each fungal strain. Compared with WT, the conidiation yield of $\Delta sntB$ on the infected host was significantly reduced ($P < 0.001$) and no AFB1 could be detected on the hosts infected by $\Delta sntB$, while in the Com-*sntB* strain, the capacity to produce conidia and AFB1 on both crop kernels recovered (Figure 2A–2C and Figure S3A). The role of *sntB* in fungal virulence to animals was also investigated. As Figure 2D–2E shown, the survival rate of silkworms injected by spores of $\Delta sntB$ strain was significantly higher than that from WT infected larvae. There was less fungal mycelium, conidia, and AFB1 production on the dead silkworms injected by $\Delta sntB$ compared to the silkworms from the WT injection groups, but when the gene was reintroduced (i.e. the Com-*sntB* group), similar to what found in the WT group, the survival rate of silkworms obviously dropped and more fungal mycelium, conidia, and AFB1 produced on the dead silkworms (Figure 2F–2H). All the above results revealed that SntB plays an essential role in virulence of *A. flavus*.

The capacity of fungal infection is closely related to secreted hydrolases, such as amylase, lipase, protease, etc. In order to explore the effect of SntB on the activity of hydrolases, the activities of amylase in the above each fungal strain were further determined. The results showed that the colonies of $\Delta sntB$ produced almost no degradation transparent circle after adding iodine solution compared with that of WT and Com-*sntB*, which indicated that the activity of α -amylase in $\Delta sntB$ strain was significantly reduced ($P < 0.001$) (Figure S3B and S3C). This suggests that SntB plays an important role in the fungal pathogenicity by changing the hydrolases activity of *A. flavus*.

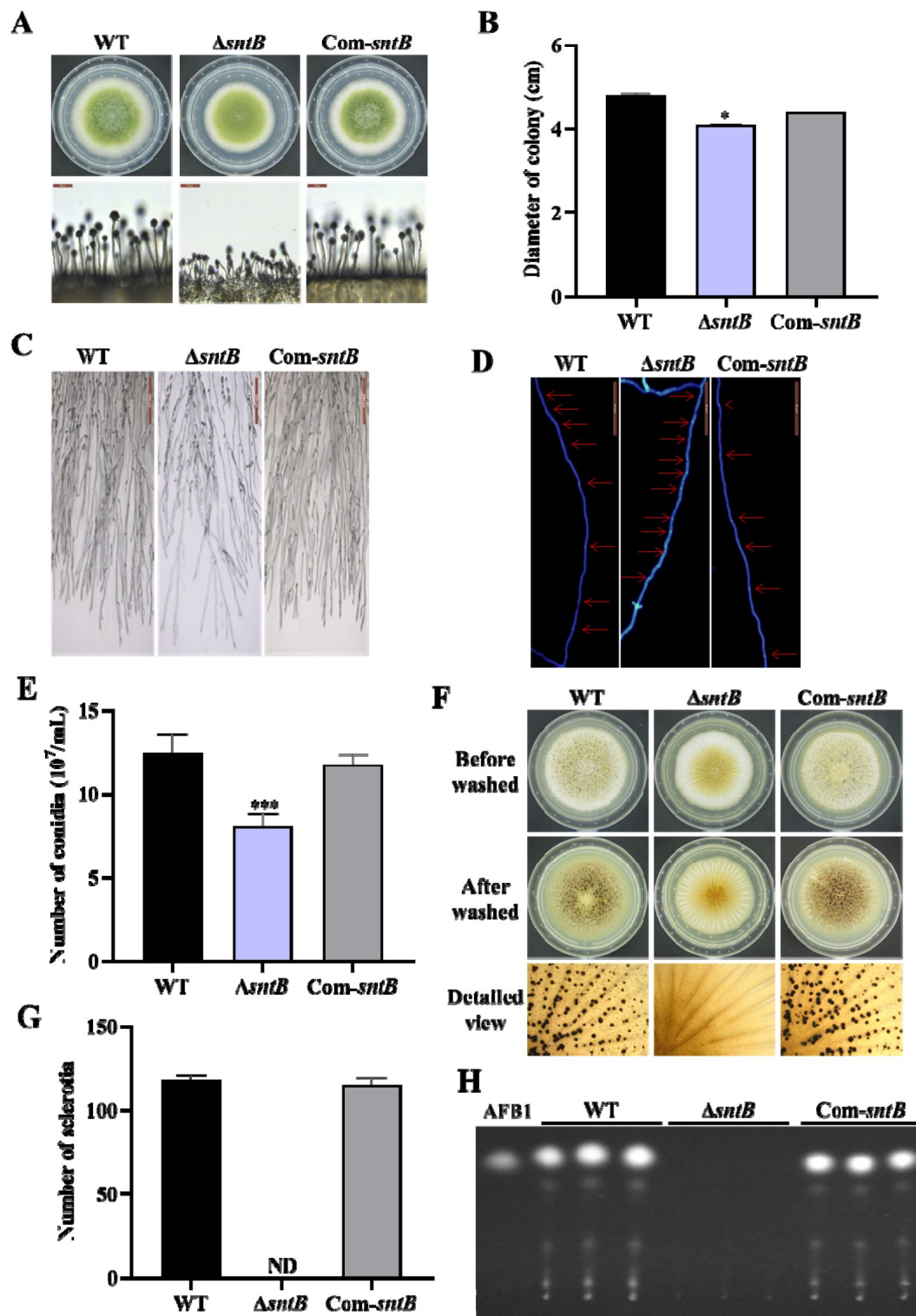


Figure 1.

The functions of SntB in *A. flavus*. (A) The colonies of WT, $\Delta sntB$, and Com-*sntB* strains grown on PDA at 37°C in dark for 4 d. (B) The colony diameter statistics of the above fungal strains. (C) Microscopic examination revealed the difference in mycelia of each fungi strain at 37L in dark, scale=200 μm . (D) Microscopic examination of the hyphal septum of each strain at 37L in dark, scale=50 μm . (E) The spore production statistics. (F) All the above fungal strains were point-inoculated on CM medium and grown for 7 d at 37L. (G) The number of sclerotia of the above fungal strains. ND=Not detectable. (H) AFB1 production of the above fungal strains was detected by TLC after the strains incubating at 29L in PDB medium for 7 d.

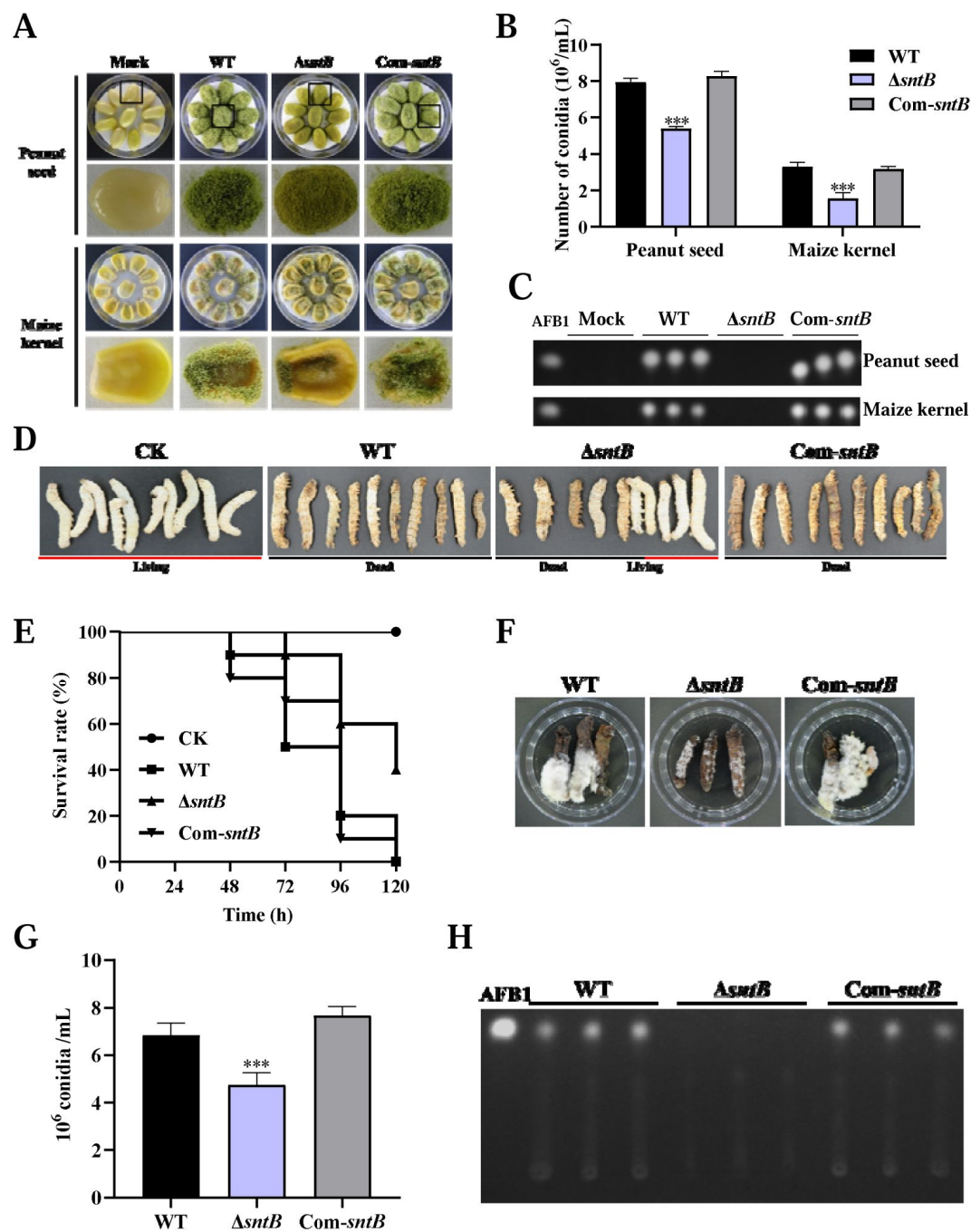


Figure 2.

The role of *SntB* on the ability of *A. flavus* to colonize host. (A) Phenotypic of corn and peanut kernels colonized by $\Delta sntB$, Com-*sntB*, and WT strains at 29°C in dark for 7 d. (B) Statistical of the number of conidia on the surface of peanut and maize kernels. (C) TLC analysis to detect the yield of AFB1 in kernels infected by the above fungal strains after 7 d incubation. (D) Photographs of the silkworms infected by the above fungal strains. (E) The survival rate of silkworms 1 week after injection of the above strains. (F) Photographs of the dead silkworms infected by *A. flavus* after 6 d incubation. (G) The spore production statistics of the above fungal strains on the dead silkworms shown in (F). (H) TLC analysis of AFB1 levels produced in infected dead silkworms in (F).

SntB chords global gene expression

To explore the downstream signaling pathways regulated by SntB, samples with three biological replicates of WT and Δ *sntB* strains were submitted for RNA-Seq. In the assay, the bases score Q30 was more than 93.19% (Table S1) and mapping ratio was from 95.17% to 95.80% (Table S2). To further confirm the quality of RNA-Seq, PCA (Principal Components Analysis) and Pearson correlation analysis were performed. Correlation analysis revealed that the samples were clustered by groups (**Figure 3A** [↗](#)). A plot of PC1 (47.70%) and PC2 (23.20%) scores showed a clear separation between the groups (**Figure 3B** [↗](#)). A total of 1,446 and 1,034 genes were significantly up- and down-regulated, respectively, in the Δ *sntB* compared to the WT strain (**Figure 3C** [↗](#) and Supplementary Table S3). GO enrichment analysis identified 93 enriched GO terms ($P < 0.05$) (**Figure 3D** [↗](#) and Supplementary Tables S4). In the biological process category, the most enriched terms were “oxidation-reduction process (GO: 0055114)”. In the molecular function category, “catalytic activity (GO: 0003824),” “oxidoreductase activity (GO: 0016491),” and “cofactor binding (GO: 0048037)” were the most significantly enriched terms. Whereas terms associated with “Set3 complex (GO: 0034967),” “mitochondrial crista junction (GO: 0044284)” and “extracellular region (GO: 0005576)” were significantly enriched in the cellular component category. Additionally, all the DEGs were mapped according to the KEGG database, and 42 significantly enriched pathways were identified ($P < 0.05$) (Supplementary Table S5). Among them, “metabolic pathways (ko01100),” “aflatoxin biosynthesis (ko00254),” and “microbial metabolism in diverse environments (ko01120)” were the most significantly enriched (**Figure 3E** [↗](#) and Supplementary Table S5).

Characterization the binding regions of SntB by chromatin immunoprecipitation sequencing (ChIP-seq)

To characterize the chromatin regions targeted by SntB, ChIP-seq studies were carried out with both HA tag fused *sntB* strain (*sntB*-HA) and WT strain. The *sntB*-HA strain was constructed by homologous recombination through fused HA to the 3' end of *sntB* (**Figure 4A** [↗](#)). In the ChIP-seq assay, more than 94.66% of bases score Q30 and above in each sample (Table S6), and reaching 52.50% to 94.48% of mapping ratio (Table S7). The principal component analysis (PCA) (**Figure S4A** [↗](#)) and heat map (**Figure S4B** [↗](#)) reflected that the quality of samples was competent for subsequent ChIP-seq analysis. There were 1,510 up-enriched differently accumulated peaks (DAPs) in *sntB*-HA fungal strain compared to the WT strain, which were distributed on the whole *A. flavus* genome (**Figure 4B** [↗](#), and Table S8). Most of the up-enriched peaks were located in the promoter (82.85%) region (**Figure 4C** [↗](#)). To determine binding regions of SntB, we used the HOMER known and *de novo* motif discovery algorithm. Motifs were sorted based on p -values and the top 5 enriched known motifs were shown in **Figure 4D** [↗](#). The results consisted of motifs derived from previously-published ChIP-Seq experiments on Cbf1, bHLHE40, NFY, Usf2, and USF1 (Table S9). However, the most enriched *de novo* motif was NFYA (1e-97) (**Figure 4E** [↗](#)). The genes of the DAPs were further subjected to GO and KEGG analysis. The most strikingly enriched GO terms in the biological process category were “cell communication (GO:0007154),” “response to stimulus (GO:0050896),” and “response to external stimulus (GO:0009605)”. Whereas terms associated with “DNA-binding transcription factor activity (GO:0003700),” “DNA-binding transcription factor activity, RNA polymerase II-specific (GO:0000981),” and “sequence-specific DNA binding (GO:0043565)” were the most significantly enriched molecular function category (**Figure 4F** [↗](#) and Table S10). And these genes were mostly enriched in “Methane metabolism” and “MAPK signaling pathway - yeast” pathways (P value < 0.05) (**Figure 4G** [↗](#) and Table S11).

Integration of the results of ChIP-seq and RNA-seq assays

After the overlapping the results from both different sequence methods (ChIP-seq and RNA-seq), 238 DEGs were found (**Figure 5A** [↗](#)). According to the GO annotation, these DEGs were significantly enriched in 8 GO terms, including “cellular response to reactive oxygen species

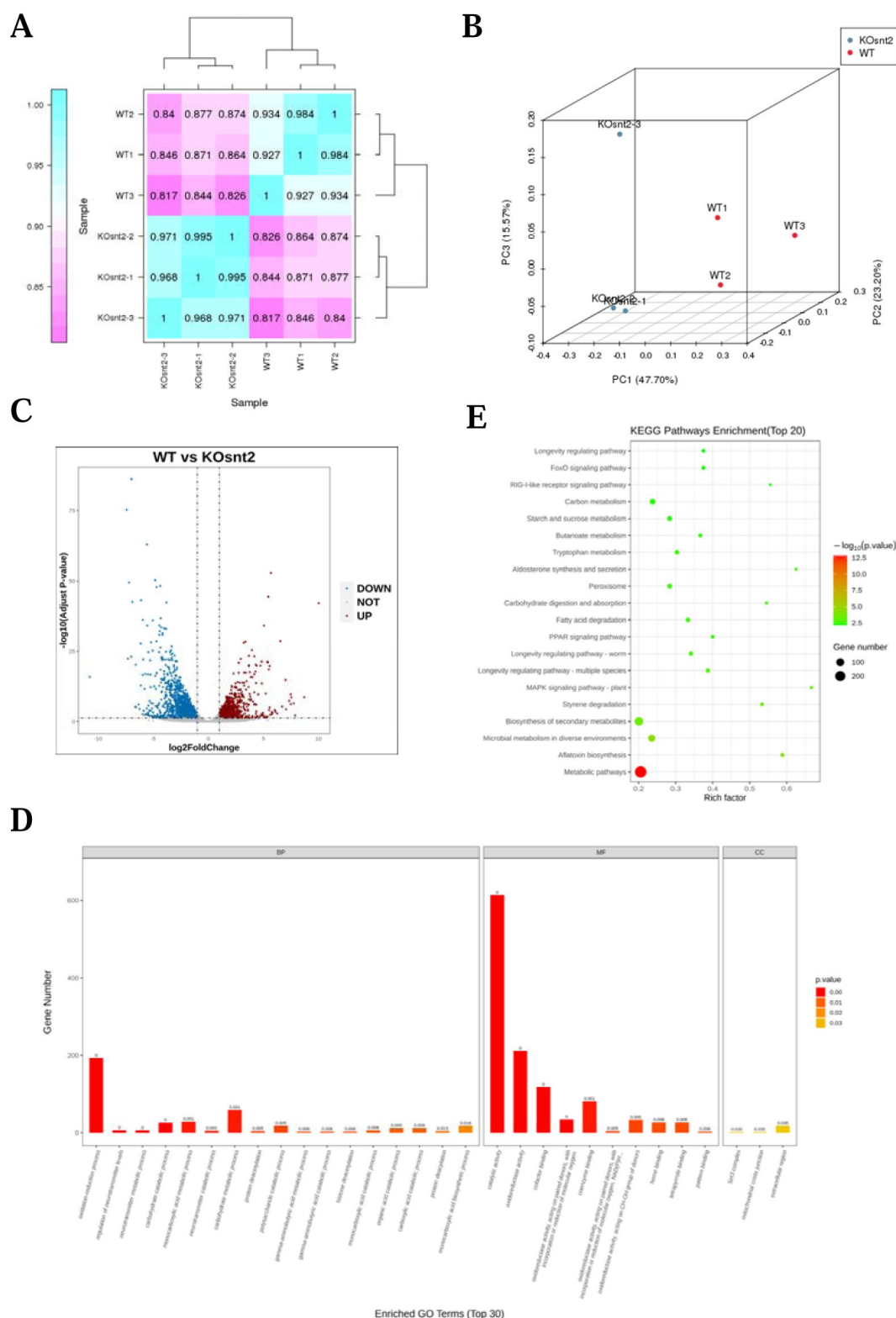


Figure 3.

SntB chords global gene expression in *A. flavus*. (A) The Pearson correlation results shown by heatmap. (B) Principal component analysis (PCA) on six fungal samples, including three Δ *sntB* (KO_{snt2}) and three WT fungal strains. (C) Volcano map reflecting the distribution of the differential expression genes. (D) Gene Ontology (GO) analyses of the differentially expressed genes. (E) Kyoto encyclopedia of genes and genomes (KEGG) analyses of the differentially expressed peaks related genes.

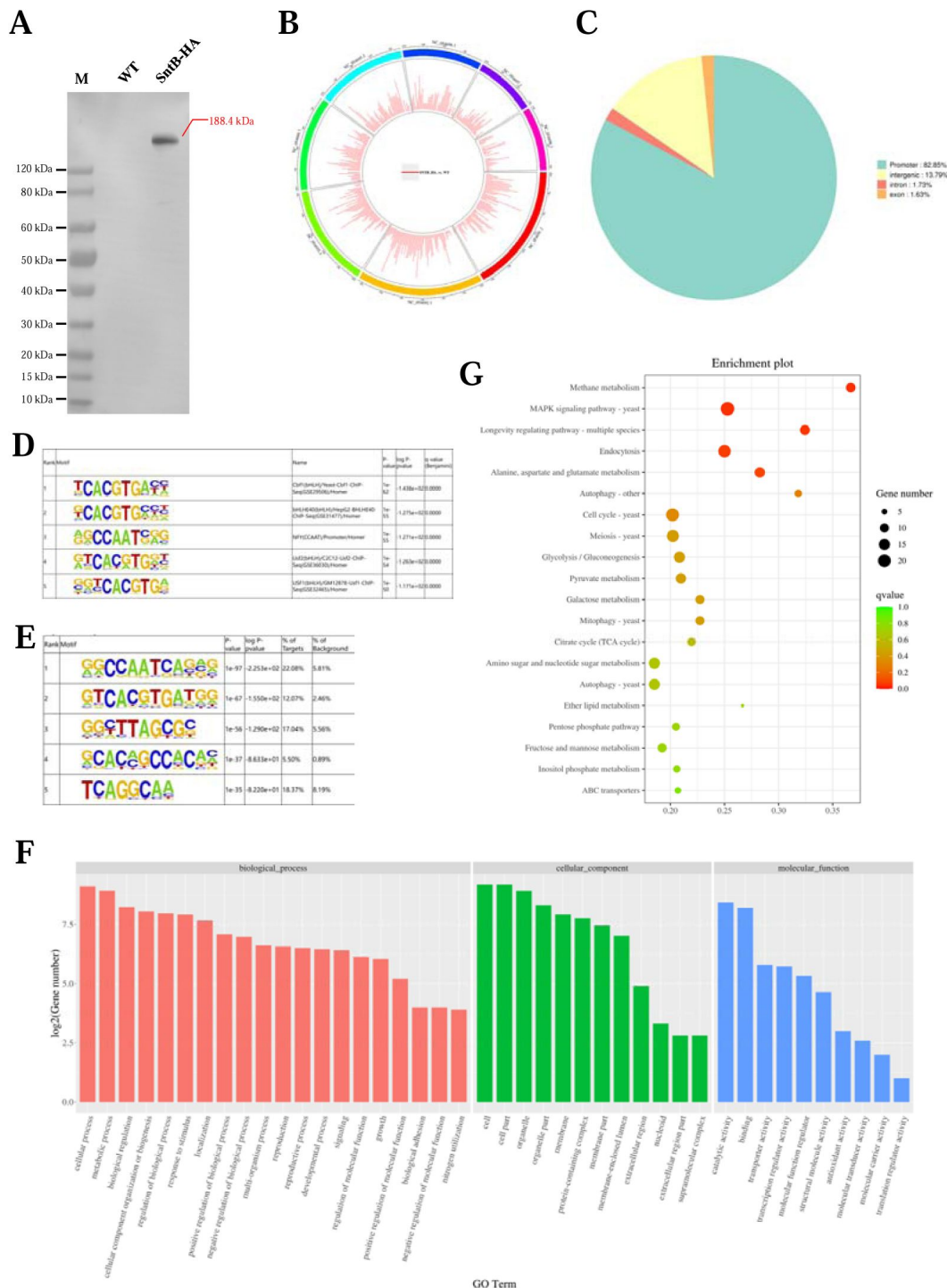


Figure 4.

Characterization the binding regions of SntB. (A) Verification of the construction of *sntB*-HA strain using western blot. M means the PAGE-MASTER Protein Standard Plus (GenScript USA, MM1397). (B) The distribution of differentially accumulated peaks on the genome. (C) Vennpie map of the differentially accumulated peaks distribution on gene functional elements. (D) Enrichment of known motifs showing the top-ranked motif logos. (E) Enrichment of *de novo* motifs showing the top-ranked motif logos. (F) GO analyses of the of the differentially accumulated peaks related genes. (G) KEGG analyses of the differently accumulated peak related genes.

(GO:0034614)”, “reactive oxygen species metabolic process (GO:0072593)”, and “cellular response to oxygen-containing compound (GO:1901701)” (**Figure 5B** [↗](#)). It was further noted that the DEGs were significantly assigned to “Carbon metabolism (afv01200)”, “Peroxisome (afv04146)”, and “Glyoxylate and dicarboxylate metabolism (afv00630)” KEGG pathways (**Figure 5C** [↗](#) and Table S12-S13). These results revealed that SntB is essential for *A. flavus* to maintain the homeostasis of intracellular reactive oxygen species. Studies had shown that SntB could response to oxidative stress in yeast [27 [↗](#)] and *Magnaporthe oryzae* [51 [↗](#)]. As **Figure 5D** [↗](#) and **Figure 5E** [↗](#) showed, due to the deletion of the *sntB* gene, $\Delta sntB$ exhibited a severe MSB sensitivity phenotype compared to that of the wild-type (WT) strain, and the phenotype recovered in the complementary strain (Com- $\Delta sntB$). The results showed that the inhibition rate of oxidant MSB to $\Delta sntB$ would be significantly enhanced with the increase of MSB concentration. These results showed that SntB deeply participates in the regulation of oxidative stress pathway. As the most abundant peroxisomal enzyme, catalases catalyze decomposition of hydrogen peroxide [52 [↗](#)]. As the most abundant peroxisomal enzyme, catalases catalyze decomposition of hydrogen peroxide [52 [↗](#)]. To further study the mechanism of SntB mediated oxidative response of *A. flavus*, the *catC* (encode a catalase) gene was selected based on the above integration results. According to the peak map, the binding region of SntB on *catC* gene was significantly enriched in *sntB*-HA strains compared to WT strain (**Figure 4F** [↗](#)). The motif in the binding region was shown in **Figure 4G** [↗](#) and the sequence was TCCGCCCG. The relative expression levels of *catC* in wild-type and $\Delta sntB$ strains under MSB treatment were measured. As **Figure 5H** [↗](#) shown, the expression level of *catC* was significantly higher in $\Delta sntB$ strain than in WT strain, it suggested that to compensate the absence of *sntB*, *catC* is up-regulated to respond the higher intracellular oxidative level. However, under the stress of oxidant MSB, the deletion of *sntB* obvious suppressed the expression level of *catC* compared to that of WT strain, which reflected that the absence of *sntB* significantly impaired the capacity of *catC* to further respond to extra external oxidative stress. These results revealed that SntB is deeply involved in CatC mediated oxidative response in *A. flavus*.

CatC is important for *A. flavus* response to oxidative stress

The functions of *catC* gene in *A. flavus* were further explored by knockout of the *catC* (**Figure S5** [↗](#)). As shown in **Figure 6A** [↗](#)-**6C** [↗](#), the diameter of $\Delta catC$ strain was significantly smaller than that of WT, and the conidia number in the $\Delta catC$ strain decreased significantly compared to that of WT. The sclerotia production of $\Delta catC$ strain was also significantly less than that of WT strain (**Figure 6D** [↗](#) and **6E** [↗](#)). In view of CatC is involved in the oxidative response pathways of *A. flavus* (**Figure 4** [↗](#)), both $\Delta catC$ and WT strains were treated by a serial concentration of MSB, and the results showed that the inhibition rates of MSB to $\Delta catC$ strain was significantly lower than that of WT group (**Figure 6F** [↗](#) and **6G** [↗](#)). Catalase is a major peroxisome protein and plays a critical role in removing peroxisome-generated reactive oxygen species. The result of fluorescence intensity of oxidant-sensitive probe dichlorodihydrofluorescein-diacetate (DCFH-DA) showed that ROS accumulation in the $\Delta catC$ strain was higher than that in the WT strain (**Figure 6G** [↗](#)). This result echoed that the deletion of *sntB* increased intracellular oxidative level and the inhibition rate of MSB, and up-regulated the expression of *catC* (**Figure 5D** [↗](#)-**5F** [↗](#)). The role of CatC in the bio-synthesis of AFB1 was also assessed (**Figure 6H** [↗](#)). The results showed that a relatively large amount of AFB1 was produced by the $\Delta catC$ strain compared to the WT. But when under the stress of MSB, AFB1 yield of the WT strain was significantly more than that of $\Delta catC$ strain. All the above results revealed that the *catC* plays an important role in SntB mediated regulation pathway on fungal morphogenesis, oxidative stress responding and AFB1 production.

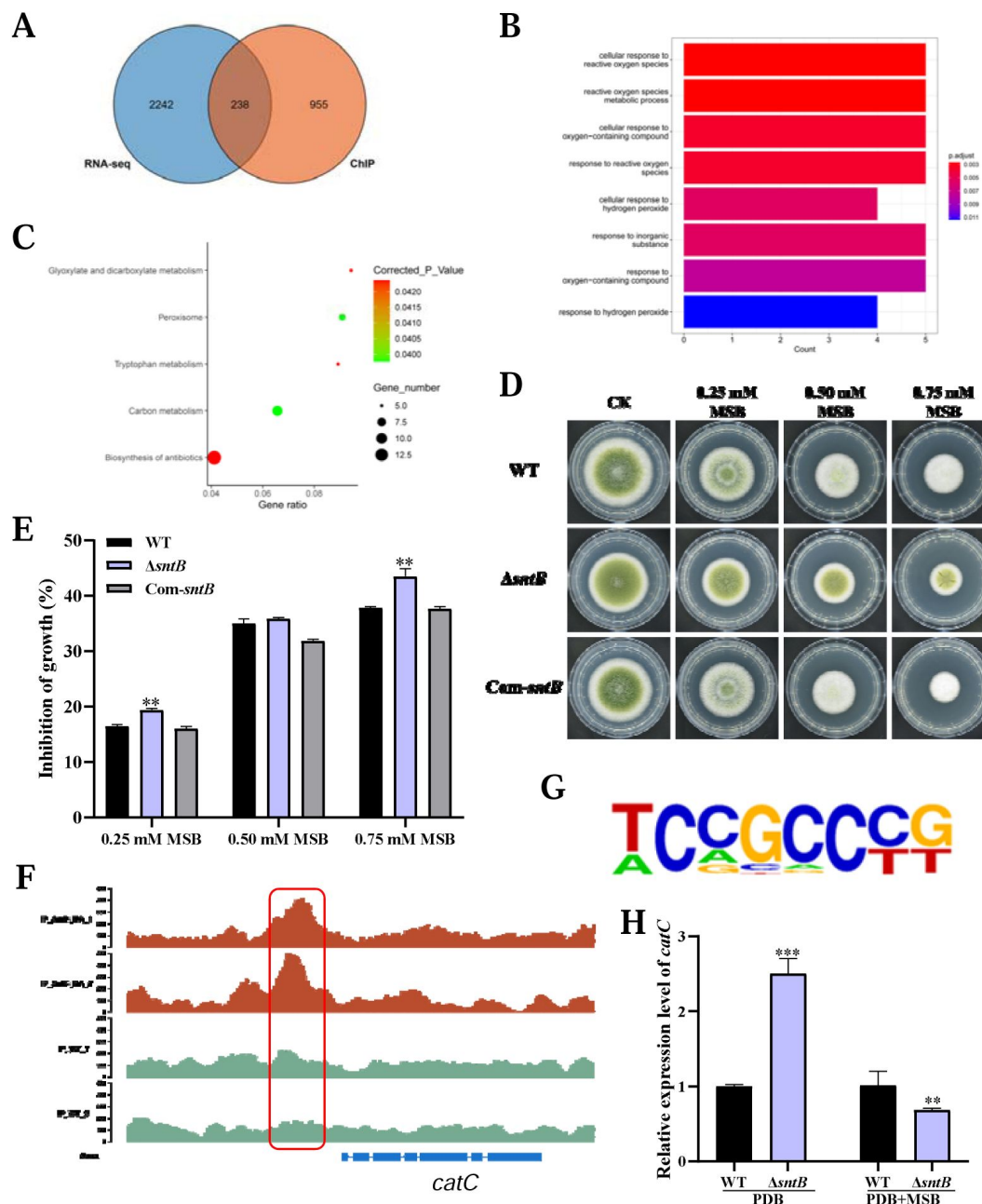


Figure 5.

Integration of the results of ChIP-seq and RNA-seq assays. (A) Venn diagrams of ChIP-seq and RNA-seq. (B) GO analyses of the common genes. (C) KEGG analyses of the common genes. (D) The phenotype of WT, $\Delta sntB$, and Com- $sntB$ strains cultured in PDA containing a series concentration of MSB for 3 d. (E) Statistical analysis of the growth inhibition rate of MSB to all the above fungal strains according to Panel D. (F) Comparison of the enrich levels of the SntB binding region of $catC$ gene between WT and $sntB$ -HA strains. (G) The motif logo in the SntB binding region of $catC$ gene. (H) The relative expression level of $sod1$ in WT and $\Delta sntB$ strains with or without MSB treatment.

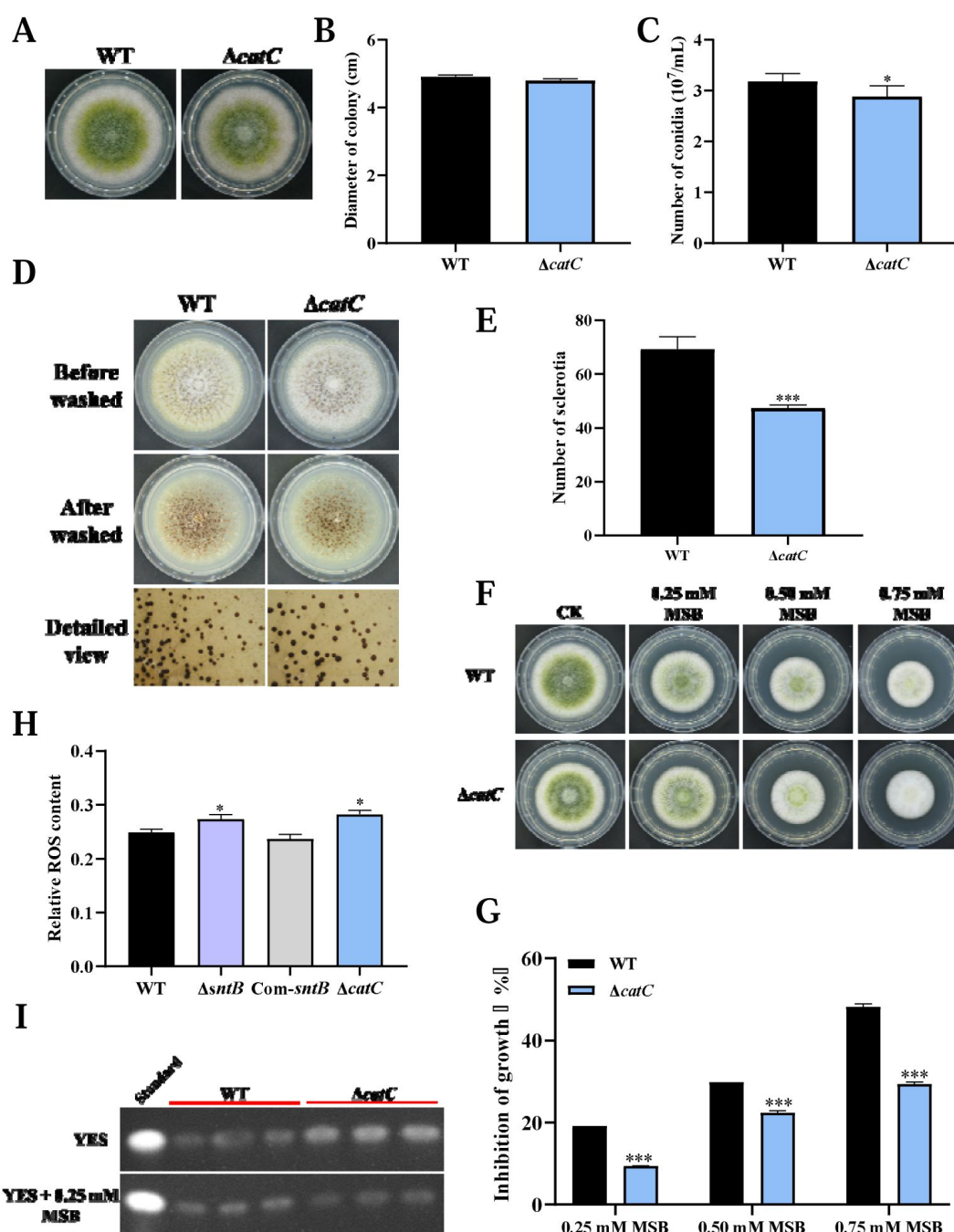


Figure 6.

The functions of *catC* in *A. flavus*. (A) The colonies of WT and $\Delta catC$ strains grown on PDA at 37°C in dark for 4 d. (B) The colony diameter statistics of the above fungal strains. (C) The spore production statistics of the above fungal strains. (D) All above fungal strains were point-inoculated on CM medium and grown for 7 d at 37L. (E) The number of sclerotia of the above fungal strains. (F) The phenotype of above strains cultured in PDA medium containing a series concentration of MSB for 3 d. (G) Statistical analysis of the growth inhibition rate of MSB to all the above fungal strains according to (F). (H) Relative ROS level in the WT, $\Delta sntB$, Com-*sntB*, and $\Delta catC$ strains. (I) AFB1 production of the above fungal strains was detected by TLC after the strains incubating at 29L in PDB medium for 7 d.

SntB regulates fungal virulence through peroxidase mediated lipolysis

Biogenesis of peroxisomes was reported to promote lipid hydrolysis, produce the production of glycerol, and further change fungal pathogenicity [53]. Due to deeply involved in oxidative response of *A. flavus*, we wondered if SntB also takes part in the regulation of the production of lipid and glycerol. As shown in Table S14, one gene (G4B84_008359) in lipase activity GO term was significantly down-regulated in Δ sntB strain, which encoded a secretory lipase belonged to the virulence factors reported in *Pseudomonas aeruginosa* [54]. The lipase activity was assayed by examining the ability to cleave glycerol tributyrin substrate [55]. The results showed that the colony diameter of Δ sntB strain on PDA medium with tributyrin were significantly smaller than that of the control, and the colony diameters of WT and Com-sntB strains on PDA medium were obviously bigger than those on 0.3% tributyrin PDA medium. The relative inhibition rate of tributyrin to the colony growth of Δ sntB strains was significantly higher than that of WT and Com-sntB strain (Figure S3D and S3E). Our previous study revealed that H3 lysine 36 trimethylation (H3K36me3) modification on the chromatin region of the sntB is regulated by AshA and SetB [34]. H3K36me3 usually promote gene transcription [56, 57]. Our study revealed the potential machinery associated with SntB mediated regulation on fungal morphogenesis, mycotoxin anabolism and fungal virulence, which lurks the axle of from SntB to fungal virulence and mycotoxin bio-synthesis through lipid catabolism (i.e. H3K36me3 modification-SntB-Peroxisomes-Lipid hydrolysis-fungal virulence and mycotoxin bio-synthesis).

Discussion

SntB is a conserved regulator in many species, including *A. nidulans* [25], *Saccharomyces cerevisiae* [27], *Schizosaccharomyces pombe* [58], *F. oxysporum* [31], *Neurospora crassa* [32], and *A. flavus* [20, 25]. SntB can regulate the production of uncharacterized secondary metabolites, including aspergillicin A1 and aspergillicin F2 [59]. MoSntB protein was required for regulation of infection-associated autophagy in *M. oryzae* [51]. In *A. flavus*, the functions of sntB gene were analyzed by both Δ sntB and overexpression sntB genetic mutants [20]. SntB deletion impaired several developmental processes, such as sclerotia formation and heterokaryon compatibility, secondary metabolite synthesis, and ability to colonize host seeds, which were consistent with our results (Figure 1 and 2). Unlike, a complementation strain was constructed in this study which further clarified and confirmed the function of sntB gene. In this study, the potential mechanism under these effects were further analyzed by detection the related transcriptional factor genes of sporulation (*steA*, *WetA*, *fluG*, and *veA*), sclerotia formation (*nsdC*, *nsdD*, and *sclR*), and the AFs synthesis related genes *aflC*, *aflD*, *aflO*, *aflP*, and *aflR* (Figure S2). In the RNA-seq data, we also found some DEGs related to AFs synthesis (*aflB*, *aflE*, *aflH*, *aflK*, *aflN*, *aflO*, *aflP*, *aflQ*, *aflR*, *aflS*, *aflV*, and *aflW*) (Figure S6). And all these genes were down-regulated, which was consistent with that the AFs production in Δ sntB was significantly decreased compared to WT and Com-sntB (Figure 1F). These results inferred that SntB regulated the morphogenesis and the production of *A. flavus* through the above canonical signal pathways.

For the process of *A. flavus* invading hosts, in view of it is a notorious pathogen for plant and animal, we established both crop and insect models, especially silkworm represented animal mode profoundly revealed the critical role of SntB in fungal virulence (Figure 2). The results of crop kernel models showed that the number of spores of Δ sntB on kernels of both peanut and maize was dramatically lower than that of strains WT and Com-sntB (Figure 2A and 2B) and almost no AFB1 was detected on maize and peanut kernels inoculated with Δ sntB, while plenty of AFB1 were detected from the kernels infected by WT fungal strain, and AFB1 biosynthesis capacity of Com-sntB strains recovered compared to the Δ sntB and WT fungal strains (Figure 2C). These

results were corroborated by previous study [20]. It was also found in this study that the survival rate of silkworms injected by spores of $\Delta sntB$ strain was significantly higher than the silkworms injected with spores from WT and Com-*sntB* fungal strains (Figure 2D and 2E). What's more, we also assayed the effect of SntB on the activity amylase, which was closely related to the capacity of fungal infection [60]. As shown in Figure S3C, after adding iodine solution, the $\Delta sntB$ strain almost did not produce a degradation transparent ring compared with wild-type and complementary strains, indicating the amylase activity of the *sntB* gene knockout strain was significantly decreased ($P < 0.001$). Our results comprehensively reveal the important function of SntB in the growth, development, secondary metabolite synthesis, and virulence of *A. flavus*.

SntB was reported as an important epigenetic reader. In *A. flavus*, SntB was reported to regulate global histone modifications (acetylation and methylation) and interact with EcoA and RpdA to form a conserved chromatin regulatory complex [20, 29]. Loss of *sntB* in *M. oryzae* also led to an increase in H3 acetylation [51]. In our RNA-seq data, we also found a set domain containing histone-lysine N-methyltransferase (Ash1, G4B84_009862) was down-regulated in $\Delta sntB$ strain compared to WT (Table S3), which was reported to regulate mycotoxin metabolism and virulence via H3K36 methylation in *A. flavus* [34]. Besides, SntB is reported to be a transcriptional regulator in *A. nidulans* [25] and *F. oxysporum* [31]. So, we use RNA-seq and ChIP-seq to study the transcriptional response of *sntB* in *A. flavus*. By integration analysis the results of ChIP-seq and RNA-seq assays, we found that the enriched GO terms and KEGG pathways of the DEGs were related to oxidative response (Figure 5A–5C). These results reflected that SntB plays an important role in fungal responding to oxidative stress, which is consistent with previous reports that SntB could response to oxidative stress in yeast [27], *F. oxysporum* [31], and *M. oryzae* [51].

As a harmful by-product of oxidative metabolism, reactive oxygen species (ROS) is unavoidable and essential for fungus development [61, 62]. ROS has also been shown to be required for aflatoxin production [63–65]. Several oxidative stress-responsive transcription factors have been identified as regulating aflatoxin production, including AtfB, AP-1, and VeA [64, 66–68]. Previous studies shown that SntB protein coordinates the transcriptional response to hydrogen peroxide-mediated oxidative stress in the yeast [26, 27] and is involved in fungal respiration and reactive oxidative stress in *F. oxysporum* and *Neurospora crassa* [31, 32]. Several GO terms (“cellular response to reactive oxygen species”, “reactive oxygen species metabolic process”, and “cellular response to oxygen-containing compound”) and KEGG pathways (Peroxisome) were enriched by the DEGs screened out by integration of ChIP-seq and RNA-seq data in this study (Figure 5). And the intracellular ROS level in the $\Delta sntB$ and $\Delta catC$ strains were significantly higher than that in WT strain (Figure 6H), which was similar to previous report on the *cat1* gene in *A. flavus* [69]. This is the first time to show that the SntB in *A. flavus* is important in oxidative stress response, through which SntB participates the regulating of aflatoxin bio-synthesis and fungal development.

Fungal defense against ROS is mediated by superoxide dismutases (SOD), catalases (CAT), and glutathione peroxidases (GPX). The effect of MSB on cellular growth and antioxidant enzyme induction in *A. flavus* was previously explored [70–72]. Once in the cell, menadione may release superoxide anion [73], which was scavenged by superoxide dismutases (SOD) and transformed into hydrogen peroxide, or react with nitric oxide to form peroxynitrite [74]. This study found that after knock out *sntB* gene, the strain growth was significantly inhibited by MSB (Figure 5F and 5G). Some genes encoded superoxide dismutases and catalases were reported to associated with AF/ST synthesis [75], including *mnSOD*, *sod1*, *sod2*, *catA*, *catB*, and *hyr1*. In our RNA-seq data, 7 related genes were screened out (Table S15), among which, bZIP transcription factor *Atf21* (G4B84_008675), *catlase C* (G4B84_009242), *catlase A* (G4B84_010740), superoxide dismutases *sod2* (G4B84_003204), peroxisomal membrane protein *PmpP24* (G4B84_001452) were up-regulated, while *catlase B* (G4B84_008381) and superoxide dismutase *sod1* (G4B84_009129) were down-regulated in $\Delta sntB$ strain, respectively.

Some studies reported the correlation between ROS formation, aflatoxin production, and antioxidant enzyme activation. Aflatoxin B1 biosynthesis and the activity of total superoxide dismutase were effectively inhibited by cinnamaldehyde, whereas the activities of catalase and glutathione peroxidase were opposite [76]. The expression of *catA*, *cat2* and *sod1*, and CAT enzymatic activity were opposite correlated to AFB1 biosynthesis under AFs inhibitor piperine treatment [77]. Deletion the gene of *sod* (GenBank accession no: CA747446) reduced AFs production [78], which was most similar to *sod2* (G4B84_003204). The mitochondria-specific *sod* and the genes *aflA*, *aflM*, and *aflP* belonging to the AFs gene cluster were reported to co-regulated [79]. Ethanol can inhibit fungal growth and AFB₁ production in *A. flavus* and enhanced levels of anti-oxidant enzymatic genes including *Cat*, *Cat1*, *Cat2*, *CatA*, and Cu, Zn superoxide dismutase gene *sod1*. All these reports indicated that the expression of anti-oxidant enzymatic genes was opposite correlated to AFB1 biosynthesis.

In our study, 7 genes related to oxidative response were obviously differentially expressed in transcriptome data (Table S15). Among these DEGs, 5 out of 7 genes were up-regulated in Δ *sntB* strain. Based on the AFs production in Δ *sntB* was significantly decreased compared to WT and Com-*sntB* (Figure 1F), the most up-regulated gene in Δ *sntB* strain, *catC* (G4B84_000242), was selected for further analysis. We found that the deletion of *sntB* significantly up-regulated the *catC* gene, however, the expression of the *catC* gene was almost undetectable under MSB treatment (Figure 5F). Results also showed that the inhibition rates of MSB to Δ *catC* strain was significantly lower than that of WT group and AFB1 yield of the Δ *catC* strain was significantly decreased than that of WT strain under the stress of MSB (Figure 6F–6H). These results indicated that SntB is profoundly involved in the *catC* mediated oxidative response.

Peroxisomes are intimately associated with the metabolism of lipid droplets [80] and the histone lysine methyltransferase ASH1 promotes peroxisome biogenesis, inhibits lipolysis, and further affects pathogenesis of *Metarhizium robertsii* [53]. Set2 histone methyltransferase family in *A. flavus*, AshA and SetB, were found to regulate mycotoxin metabolism and virulence via H3K36me3, including the chromatin region of the *sntB* [34]. By ChIP-seq and RNA-seq, SntB was found to essential for *A. flavus* to maintain the homeostasis of intracellular reactive oxygen species (Figure 5A) and several anti-oxidant enzymes were up-regulated in Δ *sntB* strain (Table S14). In addition, we also found only one down-regulated DEG (G4B84_008359) in lipase activity GO term in our RNA-seq data (Table S14), which encodes a secretory lipase and belongs to the virulence factors reported in *Pseudomonas aeruginosa* [54]. These results suggested that SntB plays a pivotal role in regulating peroxisome biogenesis to promote lipolysis involving in fungal pathogenesis.

Overall, we explored and clarified the bio-function of the SntB and found that SntB responses to oxidative stress through related oxidoreductase represented by CatC in *A. flavus* (Figure 7). Our study revealed the potential machinery associated with SntB mediated regulation on fungal morphogenesis, mycotoxin anabolism and fungal virulence, which lurks the axle of from SntB to fungal virulence and mycotoxin bio-synthesis (i.e. SntB -Peroxisomes-Lipid hydrolysis-fungal virulence and mycotoxin bio-synthesis). The work of this study provided a novel perspective for developing new prevention and control strategies against pathogenic fungi.

Compliance with ethical requirements

This article does not contain any studies with human or animal subjects.

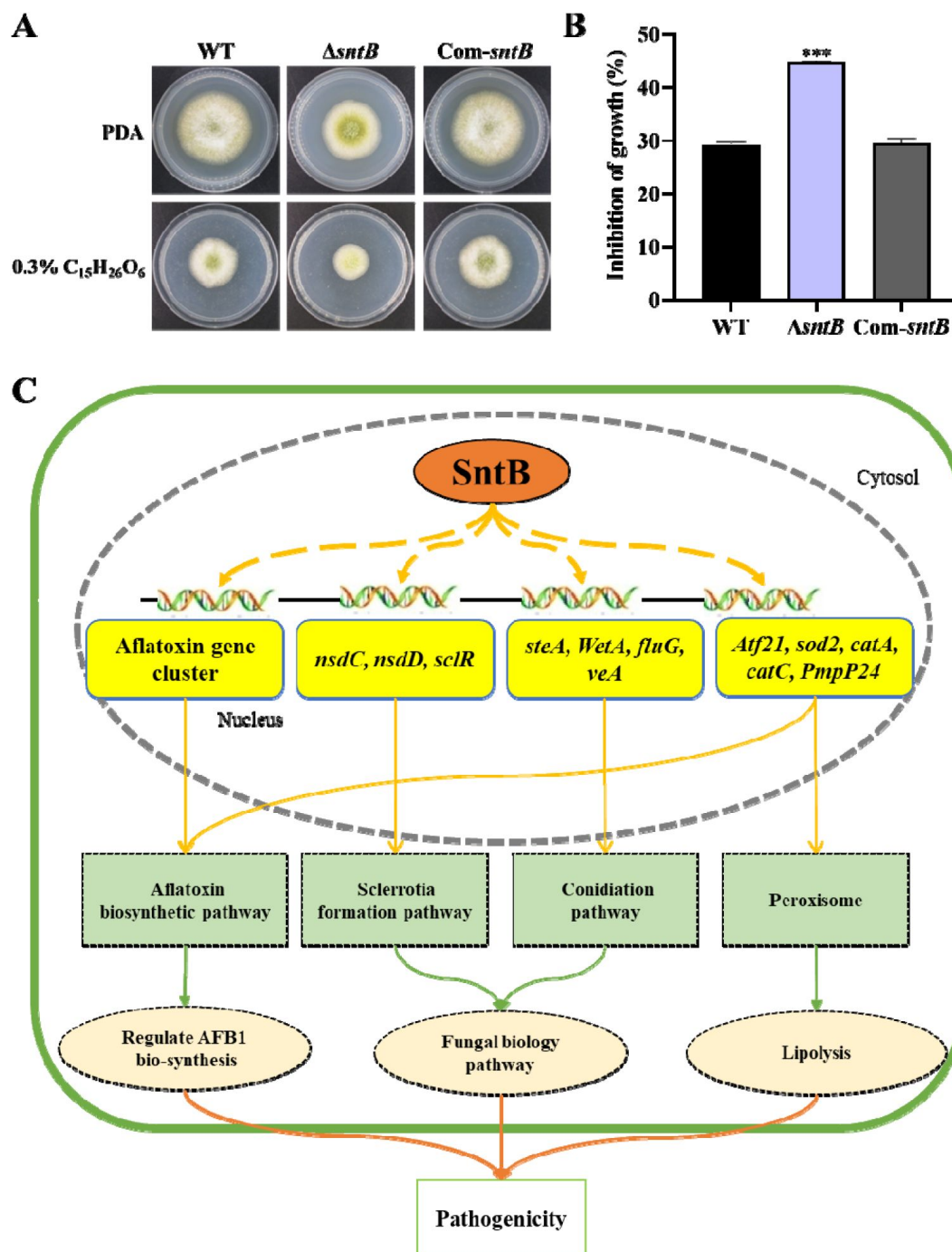


Figure 7.

SntB regulate peroxisome biogenesis, fatty acid utilization, and fungal pathogenicity in *A. flavus*. (A) The phenotype of each strain on PDA medium containing 0.3% tributyrin, (B) Statistics of inhibition rates. The asterisk *** above the bars represents significantly different ($p < 0.001$). (C) Mechanistic diagram of the bio-functions of SntB in *A. flavus*.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

All data needed to evaluate the conclusions are present in the paper and/or the Supporting Information. Raw data of the ChIP and RNA-seq were submitted to GSE247683.

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Supplementary figures

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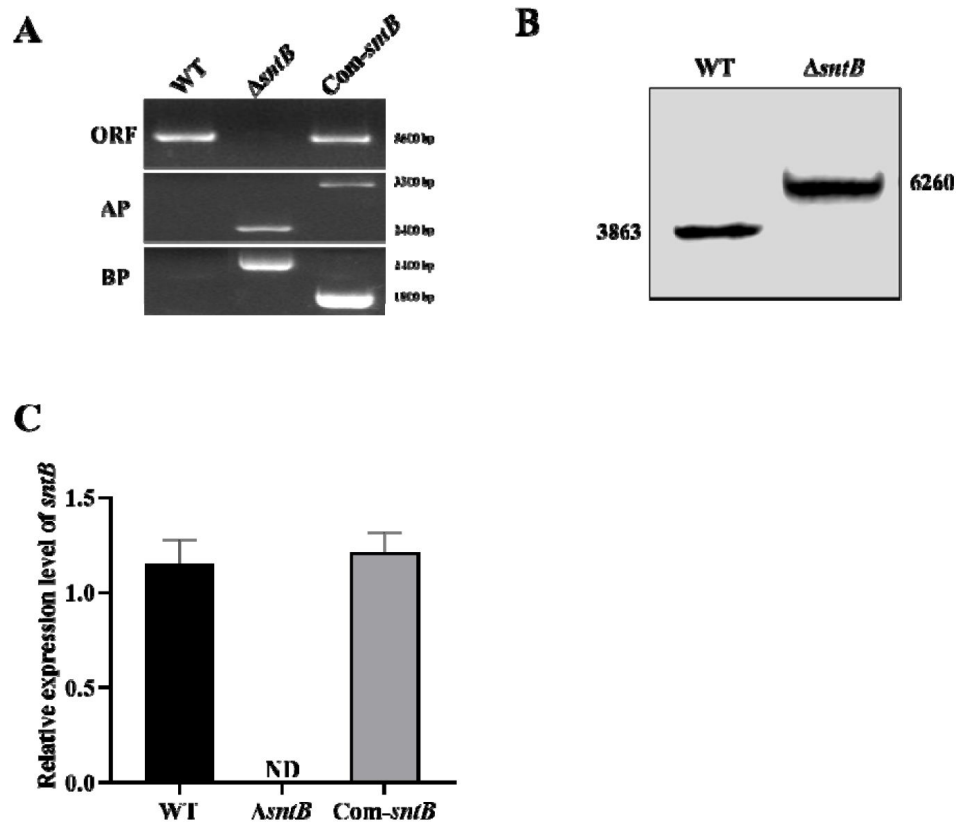


Figure S1.

The construction of mutant strains. (A) PCR verification of gDNA in WT, $\Delta sntB$ and Com-*sntB* strains ("ORF" represents the *sntB* gene fragment, "AP" represents the amplification of the fusion fragment upstream with primers *sntB*-p1 and P801, and "BP" represents the downstream of the fusion fragment from primers P1020 and *sntB*-p4). (B) Southern blot analysis result of WT, $\Delta sntB$ candidate strains (The genome DNA of each strain was digested with restrictive endonuclease EcoRI and hybridized with the probe of 1533 bp). (C) qRT-PCR verification of the expression level of the *sntB* gene in WT and *sntB* gene mutant strains.

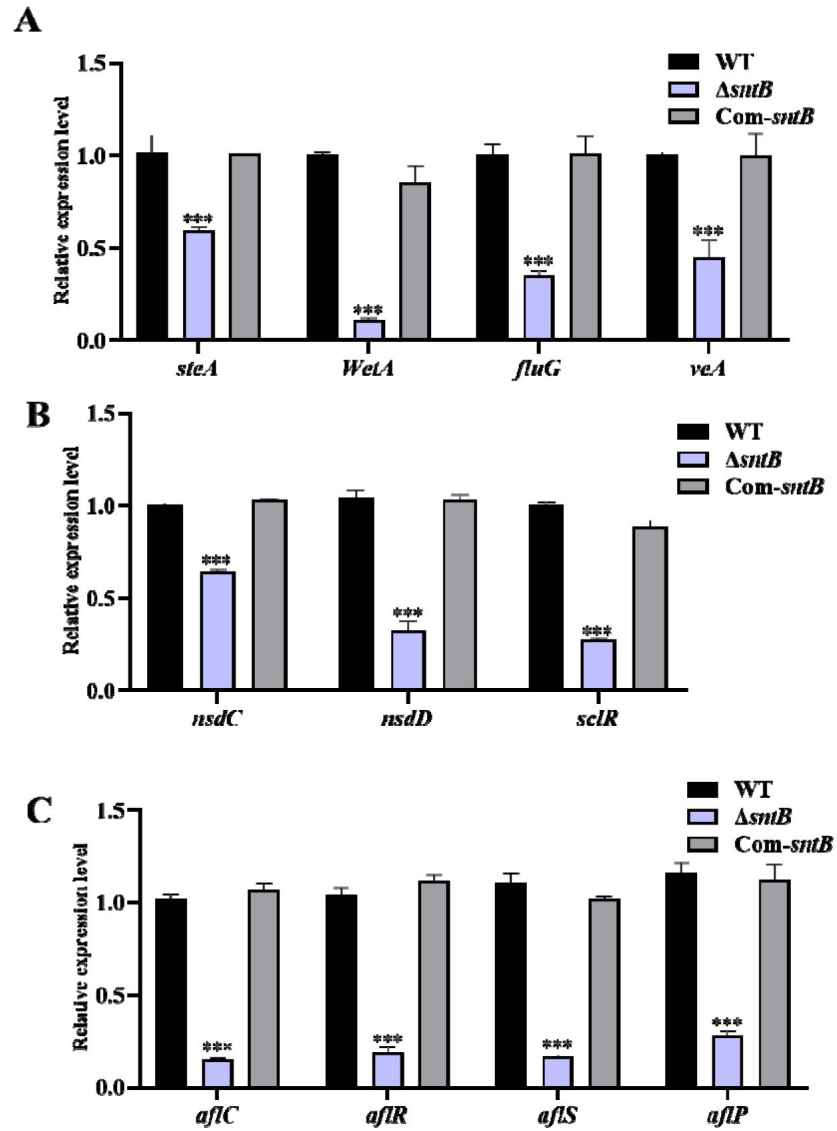


Figure S2.

The expression of genes related to sporulation, sclerotia production, and aflatoxin synthesis. (A) The expression of sporulation-related genes *steA*, *WetA*, *fluG* and *veA* in each strain at 48 h. (B) The Expression of sclerotia-associated genes *nsdC*, *nsdD*, and *sclR* in each strain at 48 h. (C) The Expression of aflatoxin-associated genes in each strain at 48 h. The asterisk *** above the bars represents significantly different ($p < 0.001$)

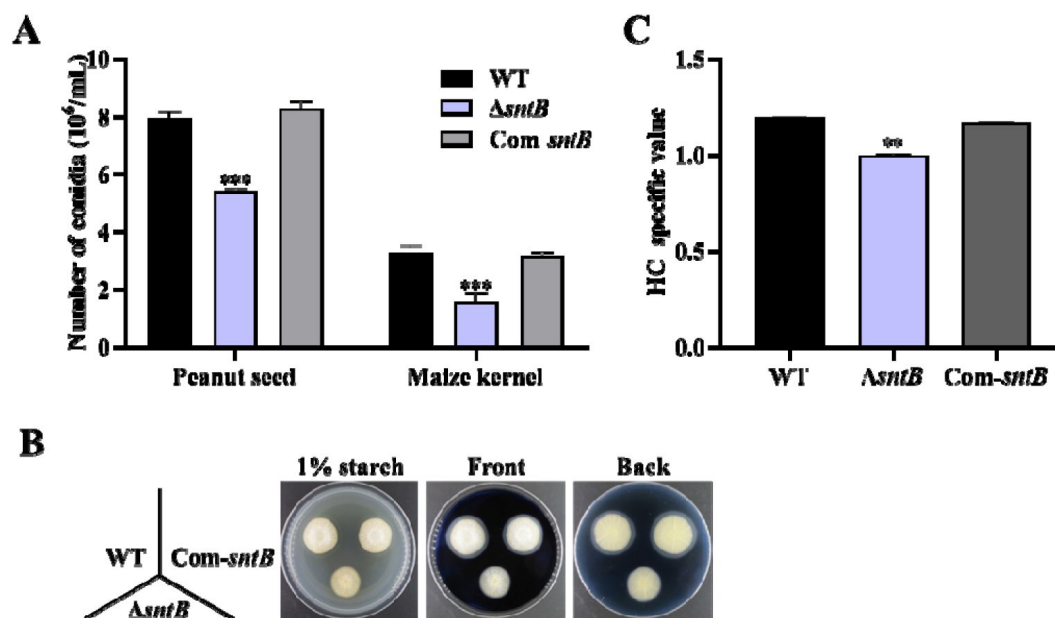


Figure S3.

The changes of number of conidia, amylase, and lipase in WT, $\Delta sntB$ and Com-*sntB* strains. (A) Statistics of the number of conidia on the corn seed. (B) The phenotype of each strain on starch screening medium supplemented with 0.1% of soluble starch in darkness at 29°C for 3 d, followed by addition of iodine solution. (C) HC value of the clear circle (Outer diameter / inner diameter) of the salient analysis of the map. The asterisk ** above the bars represents significantly different ($p < 0.01$).

Figure S4.

Sequence information of ChIP-seq. (A) Heatmap. (B) PCA. (C) Peak distribution on the genome of SNTB-HA group. (D) Peak distribution on the genome of WT group.

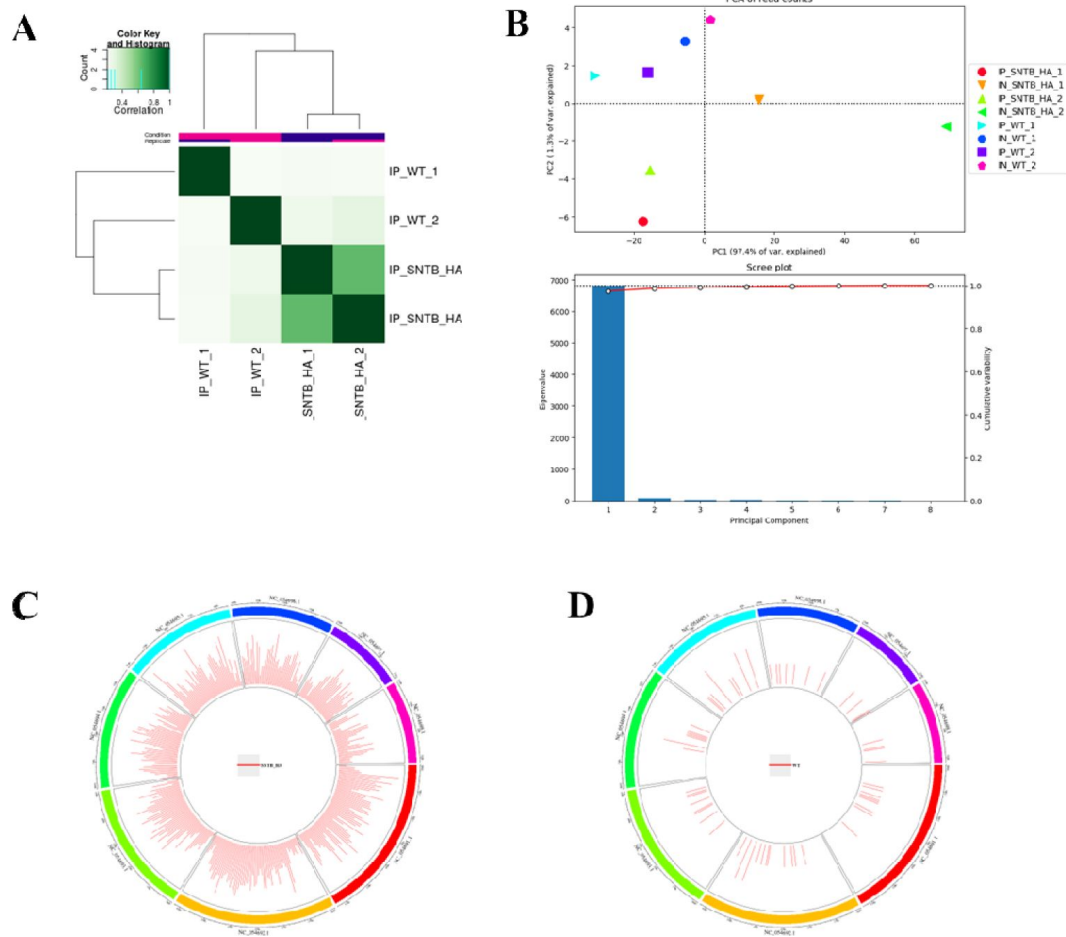
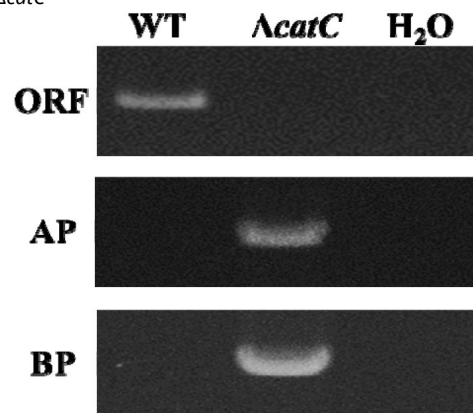


Figure S5.

PCR verification of gDNA in WT and $\Delta catC$



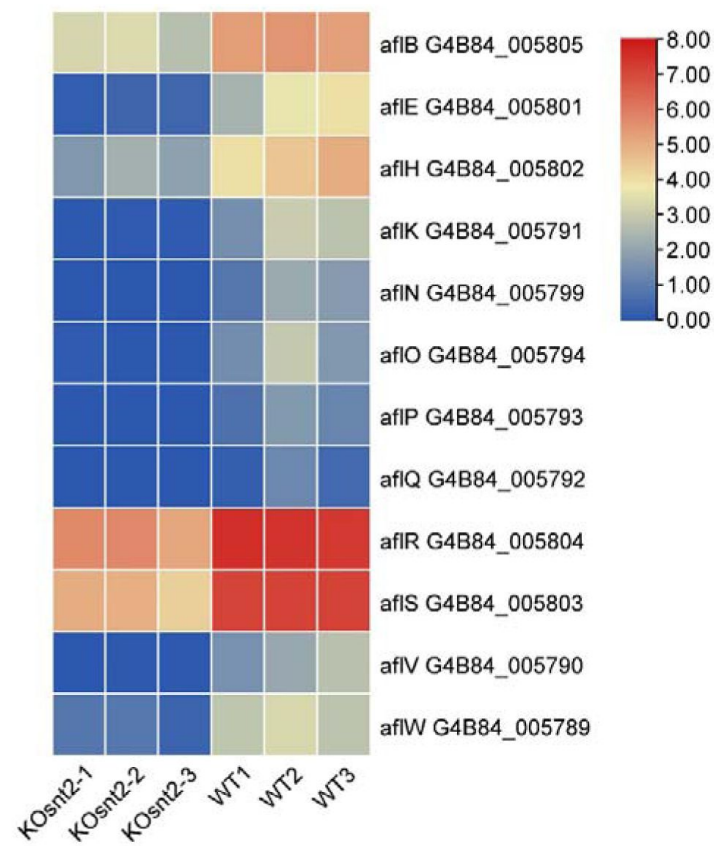


Figure S6.

Heatmap of the DEGs related to oxidative response in transcriptome data draw by TBtools.

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Editors

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Reviewer #1 (Public Review):

The study identifies the epigenetic reader SntB as a crucial transcriptional regulator of growth, development, and secondary metabolite synthesis in *Aspergillus flavus*, although the precise molecular mechanisms remain elusive. Using homologous recombination, researchers constructed sntB gene deletion (Δ sntB), complementary (Com-sntB), and HA tag-fused sntB (sntB-HA) strains. Results indicated that deletion of the sntB gene impaired mycelial growth, conidial production, sclerotia formation, aflatoxin synthesis, and host colonization compared to the wild type (WT). The defects in the Δ sntB strain were reversible in the Com-sntB strain.

Further experiments involving ChIP-seq and RNA-seq analyses of sntB-HA and WT, as well as Δ sntB and WT strains, highlighted SntB's significant role in the oxidative stress response. Analysis of the catalase-encoding catC gene, which was upregulated in the Δ sntB strain, and a secretory lipase gene, which was downregulated, underpinned the functional disruptions observed. Under oxidative stress induced by menadione sodium bisulfite (MSB), the deletion of sntB reduced catC expression significantly. Additionally, deleting the catC gene curtailed mycelial growth, conidial production, and sclerotia formation, but elevated reactive oxygen species (ROS) levels and aflatoxin production. The Δ catC strain also showed reduced susceptibility to MSB and decreased aflatoxin production compared to the WT.

This study outlines a pathway by which SntB regulates fungal morphogenesis, mycotoxin synthesis, and virulence through a sequence of H3K36me3 modification to peroxisomes and lipid hydrolysis, impacting fungal virulence and mycotoxin biosynthesis.

The authors have achieved majority of their aims at the beginning of the study, finding target genes, which led to catC mediated regulation of development, growth and aflatoxin metabolism. Overall most parts of the study is solid and clear.

<https://doi.org/10.7554/eLife.94743.2.sa1>

Reviewer #2 (Public Review):

Summary:

This work is of great significance in revealing the regulatory mechanisms of pathogenic fungi in toxin production, pathogenicity, and in its prevention and pollution control. Overall, this is generally an excellent manuscript.

Strengths:

The data in this manuscript is robust and the experiments conducted are appropriate.

Weaknesses:

- (1) The authors found that SntB played key roles in oxidative stress response of *A. flavus* by ChIP-seq and RNA sequencing. To confirm the role of SntB in oxidative stress, authors have better to measure the ROS levels in the Δ sntB and WT strains, besides the Δ catC strain.
- (2) Why the authors only studied the function of catC among the 7 genes related to oxidative response listed in Table S14.

<https://doi.org/10.7554/eLife.94743.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

The manuscript by Wu et al. explores the role of the histone reader protein SntB in Aspergillus flavus, claiming it to be a key regulator of development and aflatoxin biosynthesis. While the study incorporates various techniques, including gene deletion, ChIP-seq, and RNA-seq, several concerns and omissions in the paper raise questions about the validity and completeness of the presented findings.

(1) Omissions of Prior Work:

The authors fail to acknowledge and integrate prior research by Pfannenstiel et al. (2018) on the sntB gene in A. flavus, which covered phenotypic changes, RNA-seq data, and histone modifications. This omission raises concerns about the transparency and completeness of the current study.

The absence of reference to studies by Karahoda et al. (2022, 2023) revealing SntB's involvement in the KERS complex in A. flavus and A. nidulans is a major oversight. This raises questions about the specificity of SntB's regulatory functions, as it may be part of a larger complex. The authors should clarify why these studies were omitted and how they ensure that SntB alone, and not the entire KERS complex, is responsible for the observed effects.

We very appreciate reviewer's professional question. As reviewer mentioned, Pfannenstiel et al. (2018) reported the functions of sntB gene covered secondary metabolism, development and global histone modifications in *A. flavus* and we also cited this paper (please see reference 20). In their study, the functions of sntB gene were analyzed by both Δ sntB and overexpression sntB genetic mutants. SntB deletion impaired several developmental processes, such as sclerotia formation and heterokaryon compatibility, secondary metabolite

synthesis, and the ability to colonize host seeds, which were consistent with our results (Figure 1 and 2). Unlike, a complementation strain was constructed in our study which further clarified and confirmed the function of *sntB* gene. What's more, our main purpose is to find the downstream regulatory mechanism of SNTB, which was reported to be a transcription factor, not only as an important epigenetic reader. Please see lane 452-457 and lane 486-500.

For the function of KERS complex in *A. nidulans* (Karahoda et al., 2022), we had cited the papers, please see reference 29. For the report about the function of KERS complex in *A. flavus* (Karahoda et al., 2023), this paper was published recently. We are sorry for the omissions of this work. In our revised manuscript, we have cited this paper and compared with our work. Please see lane 97-98 and reference 30. Based solely on our experiments, we cannot confirm whether it is acting alone or in conjunction with others, what we can confirm is that SntB plays a key role in the process. And we will conduct related research in the future.

(2) Transparency and Accessibility of Data:

The lack of accessibility and visualization tools for ChIP-seq and RNA-seq data poses a challenge for independent verification and in-depth analysis. The authors should address this issue by providing more accessible data or explaining the limitations of data availability. A critical component missing from the paper is a detailed presentation of ChIP-seq data, specifically demonstrating SntB binding patterns on key promoters. This omission weakens the link between SntB and the mentioned regulatory genes. The authors should include these crucial data visualizations to strengthen their claims.

To review GEO accession GSE247683, you can go to <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE247683>, and enter the token "ipilouscnrupsl" into the box. And after our paper being published, the data will be released. For the SntB binding patterns on key promoters, we have added in the Figure 4, please see Figure 4D, 4E, 5F, 5G, and table S9.

(3) SntB Binding Sites and Consensus Sequence:

*The study mentions several genes upregulated in the *sntB* mutant without demonstrating SntB binding sites on their promoters. A detailed analysis of SntB binding maps is necessary to establish a direct link between SntB and these regulatory genes.*

Thanks for your suggestion. We have added the binding maps of SntB, please see Figure 5F, 5G; lane 362-364.

(4) Mechanistic Insight into Peroxisome Biogenesis:

*If SntB indeed regulates peroxisome biogenesis, the absence of markers for peroxisomes and the localization of peroxisomes in the *sntB* mutant vs. WT strains is a significant gap. Providing evidence for peroxisome regulation is crucial for understanding the proposed mechanism and validating the study's claims.*

Thanks for your suggestion. Catalase is ubiquitously present in aerobic organisms and plays a crucial role in mitigating oxidative stress through the scavenging of reactive oxygen species (ROS). So, we detected the ROS level in *sntB* mutant and WT strain, as well as $\Delta catC$ strain (Figure 6H).

*In summary, while the manuscript presents intriguing findings regarding SntB's role in *A. flavus*, the omissions of prior work, lack of transparency in data accessibility, and insufficient mechanistic insights call for revisions and additional experimental evidence*

to strengthen the validity and impact of the study. Addressing these concerns will enhance the manuscript's contribution to the field.

Thanks. We have revised our manuscript depending on the valuable comments provided above.

Additionally, the way the English language is used could be improved.

Thanks. We have asked a native English-writing assistant to proof read the paper and revised the grammar errors and typos and improve the readability and quality of the manuscript.

Reviewer #2 (Public Review):

Summary:

This work is of great significance in revealing the regulatory mechanisms of pathogenic fungi in toxin production, pathogenicity, and in its prevention and pollution control. Overall, this is generally an excellent manuscript.

Strengths:

The data in this manuscript is robust and the experiments conducted are appropriate.

Weaknesses:

(1) The authors found that SntB played key roles in the oxidative stress response of A. flavus by ChIP-seq and RNA sequencing. To confirm the role of SntB in oxidative stress, the authors have to better measure the ROS levels in the Δ sntB and WT strains, besides the Δ catC strain.

Thanks for your suggestion. We have supplemented the relevant experiments and the results were shown in Figure 6G and lane 185-192 and 395-398.

(2) Why did the authors only study the function of catC among the 7 genes related to an oxidative response listed in Table S14?

The function of some genes in Table S15 (Table S14 in old version of our manuscript) had been studied, such as cat1 [1]. In this study, we just choose catC for further validation, which was the most up-regulated gene in Δ sntB strain. The others may also have important roles in SntB triggered antioxidant pathways to regulate development and aflatoxin biosynthesis in *A. flavus*. We will focus on this in the following work.

(1) Zhu Z., Yang M., Bai Y., Ge F., Wang S. Antioxidant-related catalase CTA1 regulates development, aflatoxin biosynthesis, and virulence in pathogenic fungus *Aspergillus flavus* [J]. Environ Microbiol, 2020, 22(7): 2792-2810.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Line 52: Change "shad light" to "shed light"

Thanks. We have revised. Please see lane 50.

Line 62: Change "has" to "have" to match the plural noun "aflatoxins."

Original: "Aflatoxins produced by A. flavus has strong toxicity..."

Suggested: "Aflatoxins produced by A. flavus have strong toxicity..."

Thanks. We have revised it. Please see lane 62.

Line 79: Consider rephrasing for clarity.

Original: "...which may result in the modulation of the expression of genes involved in toxin production [15-17]."

Thanks. We have revised. Please see lane 77-80.

Line 105: Add a comma after "host strain."

Original: "A. flavus $\Delta ku70 \Delta pyrG$ was used as a host strain for genetic manipulations."

Suggested: "A. flavus $\Delta ku70 \Delta pyrG$ was used as a host strain, for genetic manipulations."

Thanks. We have revised it. Please see lane 107.

Line 113, Table 1: Remove the extra "r" in "from" in the Source column.

Original: "Kindly presented form Prof. Chang[1]"

Suggested: "Kindly presented from Prof. Chang[1]"

Thanks. We have revised it. Please see Table 1.

Line 140: Typo - Change "reaches" to "reach."

Original: "when silkworm larva reaches about 1 g in weight."

Suggested: "when silkworm larvae reach about 1 g in weight."

Thanks. We have revised it. Please see lane 141.

Line 158: Typo - Change "pervious" to "previous."

Original: "Data processing was according pervious study [39]."

Suggested: "Data processing was according to a previous study [39]."

Thanks. We have revised it. Please see lane 150.

Line 138 The animal invasion assay using silkworms was conducted according to a previous study.

Change "according" to "conducted according to" for clarity.

Thanks. We have revised it. Please see lane 139.

Line 148 Was carried out by APPLIED PROTEIN TECHNOLOGY, Shanghai (www.aptbiootech.com).

Change "TECHNOLOY" corrected to "TECHNOLOGY."

Thanks. We have revised it. Please see lane 149.

Line 148 Data processing was conducted according to a previous study [39].

Change "according to" to "conducted according to" for clarity.

Thanks. We have revised it. Please see lane 139.

Line 429 *Schizosaccharomyces pombe*, Correct the spelling to "*Schizosaccharomyces pombe* [55]."

Thanks. We have revised it. Please see lane 448.

Reviewer #2 (Recommendations For The Authors):

(1) The resolution of the words written in Figures 3 and 4 is not clear (or high) enough.

Thanks. We have revised them. Please see Figures 3 and 4.

(2) Which kind of protein marker (protein ladder) was used in Figure 4A, you should mark out the size of the related protein.

Thanks. We have revised. Please see Figure 4A and lane 332-333.

(3) Latin names do not necessarily need to be written in full when they are not the first time used in the text.

Thanks. We have revised them throughout the manuscript.

(4) The complementary strain of *sntB* was labeled as *sntB-C* in Figure 2B, while in other figures was *Com-sntB*. You should correct all related problems.

Thanks. We have revised it. Please see Figure 2B.

(5) What is the meaning of "1" in Table 1?

Thanks. The meaning of "1" in Table 1 was a citation. We have revised. Please see Table 1.

<https://doi.org/10.7554/eLife.94743.2.sa3>